COMICCRAFT – AI COMIC STORY CREATOR USING GEMINI MODELS

2. Requirement Analysis Phase

2.1 Introduction

Requirement Analysis is an important phase of the ComicCraft project. It identifies the functional requirements, non-functional requirements, hardware requirements, software requirements, and user requirements needed to develop the application successfully.

2.2 Functional Requirements

The ComicCraft application should provide the following functions:

- Allow users to enter a story prompt.
- Allow users to provide a character name.
- Allow users to specify the story setting.
- Allow users to select the story tone.
- Allow users to select an art style.
- Generate a comic story using Google Gemini models.
- Generate narration and dialogues.
- Generate image prompts for comic scenes.
- Generate images for the comic panels.
- Display the generated comic to the user.
- Allow the completed comic to be exported as a PDF.

2.3 Non-Functional Requirements

The application should be:

- Easy to use.
- User-friendly.
- Reliable.
- Responsive.
- Maintainable.
- Secure.
- Compatible with modern web browsers.

2.4 Hardware Requirements

The basic hardware requirements are:

- Laptop or desktop computer.

- Minimum 4 GB RAM.
- Keyboard and mouse.
- Internet connection.
- Sufficient storage space.

2.5 Software Requirements

The major software and technologies required are:

- Python
- FastAPI
- HTML
- CSS
- JavaScript
- Google Gemini
- Hugging Face
- Jinja2
- Pillow
- FPDF2
- Pytest
- Visual Studio Code

2.6 User Input Requirements

The user provides the following information:

1. Story Prompt
2. Character Name
3. Setting
4. Tone
5. Art Style

These inputs are processed by the backend and provided to the AI services.

2.7 Output Requirements

The system should generate:

- Story outline
- Narration
- Dialogues
- Image prompts
- Comic images
- Comic panels
- Final comic preview
- PDF output

2.8 Conclusion

The Requirement Analysis Phase defines the necessary features, technologies, inputs, and outputs of ComicCraft. These requirements provide a clear foundation for designing and developing the application.